

## IIT JAM 2013 Official Paper

Category: Classic Era (2005–2014) • Official Mock Test Series

TOTAL QUESTIONS

10

TOTAL MARKS

20 M

TEST DURATION

40 Min

PAPER PATTERN

MCQ

| Section / Type        |  | Questions | Marking Scheme | Negative Marking  | Format / Details      |
|-----------------------|--|-----------|----------------|-------------------|-----------------------|
| Multiple Choice (MCQ) |  | 10        | +1 / +2 Marks  | −1/3 / −2/3 Marks | 4 Options (1 Correct) |

## IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at [vaibhavgeometer.github.io](https://vaibhavgeometer.github.io).

*IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer*

Question 1 (JAM 2013 • Q1 • ID: JAM\_2013\_Q1)

MCQ

+2 M

Neg: -0.67

- Q.1 Let  $A = \begin{pmatrix} 1 & 1 & 1 \\ 3 & -1 & 1 \\ 1 & 5 & 3 \end{pmatrix}$  and  $V$  be the vector space of all  $X \in \mathbb{R}^3$  such that  $AX = 0$ . Then  $\dim(V)$  is
- (A) 0 (B) 1 (C) 2 (D) 3

Question 2 (JAM 2013 • Q2 • ID: JAM\_2013\_Q2)

MCQ

+2 M

Neg: -0.67

- Q.2 The value of  $n$  for which the divergence of the function  $\vec{F} = \frac{\vec{r}}{|\vec{r}|^n}$ ,  $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ ,  $|\vec{r}| \neq 0$ , vanishes is
- (A) 1 (B) -1 (C) 3 (D) -3

Question 3 (JAM 2013 • Q3 • ID: JAM\_2013\_Q3)

MCQ

+2 M

Neg: -0.67

- Q.3 Let  $A$  and  $B$  be subsets of  $\mathbb{R}$ . Which of the following is **NOT** necessarily true?
- (A)  $(A \cap B)^o \subseteq A^o \cap B^o$  (B)  $A^o \cup B^o \subseteq (A \cup B)^o$   
 (C)  $\overline{A \cup B} \subseteq \overline{A} \cup \overline{B}$  (D)  $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$

Question 4 (JAM 2013 • Q4 • ID: JAM\_2013\_Q4)

MCQ

+2 M

Neg: -0.67

- Q.4 Let  $[x]$  denote the greatest integer function of  $x$ . The value of  $\alpha$  for which the function
- $$f(x) = \begin{cases} \frac{\sin[-x^2]}{[-x^2]}, & x \neq 0 \\ \alpha, & x = 0 \end{cases}$$
- is continuous at  $x=0$  is
- (A) 0 (B)  $\sin(-1)$  (C)  $\sin 1$  (D) 1

Question 5 (JAM 2013 • Q5 • ID: JAM\_2013\_Q5)

MCQ

+2 M

Neg: -0.67

Let the function  $f(x)$  be defined by

$$f(x) = \begin{cases} e^x, & x \text{ is rational} \\ e^{1-x}, & x \text{ is irrational} \end{cases}$$

for  $x$  in  $(0, 1)$ . Then

- (A)  $f$  is continuous at every point in  $(0, 1)$
- (B)  $f$  is discontinuous at every point in  $(0, 1)$
- (C)  $f$  is discontinuous only at one point in  $(0, 1)$
- (D)  $f$  is continuous only at one point in  $(0, 1)$

Question 6 (JAM 2013 • Q6 • ID: JAM\_2013\_Q6)

MCQ

+2 M

Neg: -0.67

The value of the integral

$$\iint_D \sqrt{x^2 + y^2} \, dx \, dy, \quad D = \{(x, y) \in \mathbb{R}^2 : x \leq x^2 + y^2 \leq 2x\}$$

is

- (A) 0
- (B)  $\frac{7}{9}$
- (C)  $\frac{14}{9}$
- (D)  $\frac{28}{9}$

Question 7 (JAM 2013 • Q7 • ID: JAM\_2013\_Q7)

MCQ

+2 M

Neg: -0.67

Let

$$x_n = \left(1 - \frac{1}{3}\right)^2 \left(1 - \frac{1}{6}\right)^2 \left(1 - \frac{1}{10}\right)^2 \cdots \left(1 - \frac{1}{\frac{n(n+1)}{2}}\right)^2, \quad n \geq 2.$$

Then  $\lim_{n \rightarrow \infty} x_n$  is

- (A)  $\frac{1}{3}$
- (B)  $\frac{1}{9}$
- (C)  $\frac{1}{81}$
- (D) 0

Question 8 (JAM 2013 • Q8 • ID: JAM\_2013\_Q8)

MCQ

+2 M

Neg: -0.67

Let  $p$  be a prime number. Let  $G$  be the group of all  $2 \times 2$  matrices over  $\mathbb{Z}_p$  with determinant 1 under matrix multiplication. Then the order of  $G$  is

- (A)  $(p-1)p(p+1)$  (B)  $p^2(p-1)$   
 (C)  $p^3$  (D)  $p^2(p-1)+p$

Question 9 (JAM 2013 • Q9 • ID: JAM\_2013\_Q9)

MCQ

+2 M

Neg: -0.67

Q.9 Let  $V$  be the vector space of all  $2 \times 2$  matrices over  $\mathbb{R}$ . Consider the subspaces

$$W_1 = \left\{ \begin{pmatrix} a & -a \\ c & d \end{pmatrix} : a, c, d \in \mathbb{R} \right\} \text{ and } W_2 = \left\{ \begin{pmatrix} a & b \\ -a & d \end{pmatrix} : a, b, d \in \mathbb{R} \right\}.$$

If  $m = \dim(W_1 \cap W_2)$  and  $n = \dim(W_1 + W_2)$ , then the pair  $(m, n)$  is

- (A)  $(2, 3)$  (B)  $(2, 4)$  (C)  $(3, 4)$  (D)  $(1, 3)$

Question 10 (JAM 2013 • Q10 • ID: JAM\_2013\_Q10)

MCQ

+2 M

Neg: -0.67

Q.10 Let  $\mathcal{P}_n$  be the real vector space of all polynomials of degree at most  $n$ . Let  $D: \mathcal{P}_n \rightarrow \mathcal{P}_{n-1}$  and  $T: \mathcal{P}_n \rightarrow \mathcal{P}_{n+1}$  be the linear transformations defined by

$$D(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = a_1 + 2a_2x + \dots + na_nx^{n-1},$$

$$T(a_0 + a_1x + a_2x^2 + \dots + a_nx^n) = a_0x + a_1x^2 + a_2x^3 + \dots + a_nx^{n+1},$$

respectively. If  $A$  is the matrix representation of the transformation  $DT - TD: \mathcal{P}_n \rightarrow \mathcal{P}_n$  with respect to the standard basis of  $\mathcal{P}_n$ , then the trace of  $A$  is

- (A)  $-n$  (B)  $n$  (C)  $n+1$  (D)  $-(n+1)$

## OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2013 Official Paper • All Official Answers & Acceptance Ranges

| Q. No. | Type | Marks | Official Key |
|--------|------|-------|--------------|
| 1      | MCQ  | +2    | B            |
| 2      | MCQ  | +2    | C            |
| 3      | MCQ  | +2    | D            |
| 4      | MCQ  | +2    | C            |
| 5      | MCQ  | +2    | D            |
| 6      | MCQ  | +2    | D            |
| 7      | MCQ  | +2    | B            |
| 8      | MCQ  | +2    | A            |
| 9      | MCQ  | +2    | B            |
| 10     | MCQ  | +2    | C            |

### SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ( $-1/3$  for 1-mark questions,  $-2/3$  for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g.,  $[a, b]$ ) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.